

El Gamal DSA

Module V

Agent X



Spy

Agent Y



public area

Agent X



p is prime number,
 g is generator.
 g is the primitive root of p



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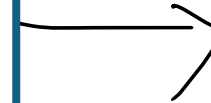
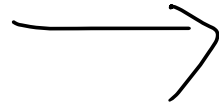
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E-Gamal DSA

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 $2 \leq d \leq p-2$.
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$e = g^d \bmod p$
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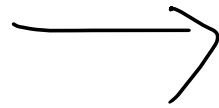


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 $m = H(M) = H(4) = 14$



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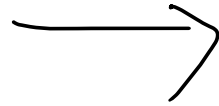


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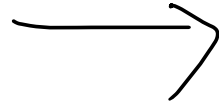


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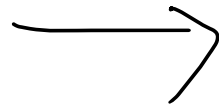
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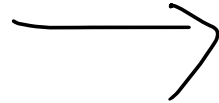


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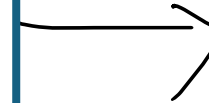


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$M=4$ **message**
 $Y1=3$
 $Y2=4$ } **signature**



Agent Y



E-Gamal DSA

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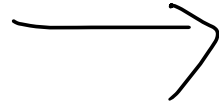


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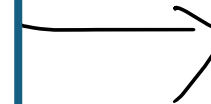
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Agent Y



$$V1 = g^m \text{ mod } p$$



E-Gamal DSA

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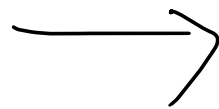


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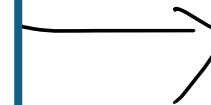
public area

Agent Y



$$V1 = g^m \text{ mod } p$$

$$V1 = 10^{14} \text{ mod } 19 = 16$$



E-Gamal DSA

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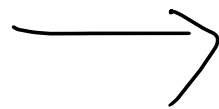


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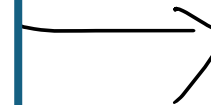
Agent Y



$$V1 = g^m \bmod p$$

$$V1 = 10^{14} \bmod 19 = 16$$

$$V2 = (e)^{Y1} (Y1)^{Y2} \bmod p$$



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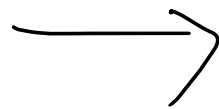


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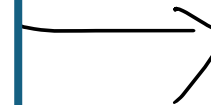


$$V1 = g^m \mod p$$

$$V1 = 10^{14} \mod 19 = 16$$

$$V2 = (e)^{Y1} (Y1)^{Y2} \mod p$$

$$V2 = (4)^3 (3)^4 \mod 19$$



E-Gamal DSA

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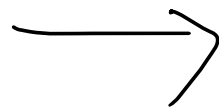


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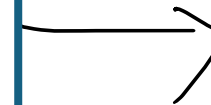
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$$V1 = 10^{14} \mod 19 = 16$$

$$V2 = (e)^{Y1} (Y1)^{Y2} \mod p$$

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$$V2 = 5184 \mod 19 = 16$$



E-Gamal DSA

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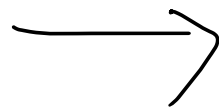
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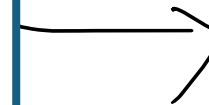
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$$V2 = (4)^3 (3)^4 \bmod 19$$

$$V2 = 5184 \bmod 19 = 16$$

Valid signature if $V1=V2$



E-Gamal DSA

El-Gamal DSA cntd..

- As with ElGamal encryption, the global elements of **ElGamal digital signature** are a prime number p and g , which is a primitive root of p . User A generates a private/public key pair as follows
- Generate a random integer d such that $2 \leq d \leq p-2$
- Compute **$e = g^d \bmod p$**
- Publish public key $\{p, g, e\}$
- Private Key is d

El-Gamal DSA cntd..

To sign a message M , user A first computes the hash $m = H(M)$, such that m is an integer in the range $0 \leq m \leq p-1$. A then forms a digital signature as follows.

- Choose a random integer such that $1 \leq k \leq p-1$
- and $\gcd(k, p-1) = 1$
- Compute **$Y1 = g^k \bmod p$**
- Compute **$K^{-1} \bmod (p - 1)$**
- $Y2 = K^{-1} (m - dY1) \bmod (p - 1)$
- Signature Consists of Pairs $(Y1, Y2)$

El-Gamal DSA cntd..

Any user B can verify the signature as follows

- Compute $V1 = g^m \bmod p$
- Compute $V2 = (e)^{Y1} (Y1)^{Y2} \bmod p$
- If $V1=V2$ then Signature is valid